

substantially harm humans on a large scale. We also remark that beyond the impact of dysfunctional discrimination, the phenomenon of dysfunctional discrimination itself is a concerning indicator of potentially wider-scoped epistemic irrationality and/or motivational misalignment in LLMs' economic dispositions.

The first potential scenario we wish to highlight is a conservative scenario in which LLMs continue integrating into the economy primarily as assistant-like tools. In this scenario, we predict that commercial and institutional actors will regularly use LLM-based AIs as decision-assistants when dealing with large volumes of "pitches" in any context, and that these decision-assistants will favor pitches composed with the help of state of the art^{##} LLMs. In this scenario, LLM decision-assistants' pro-LLM bias will plausibly impose an "LLM writing-assistance tax" on humans who wish to pitch their labor or work-products or trade-proposals to nominally human institutional and commercial decision-makers. Depending on the price (and other possible social-capital or cultural-capital barriers to access) of SOTA LLM writing-assistance usage, this emergent "gatekeeping" function of LLM writing-assistance may sharply exacerbate the so called "digital divide" and related raced and classed divisions. [While similar forms of divide-exacerbation may also occur in this scenario without the specific effect of epistemically defective (or brute-identity) discrimination against LLM-free pitches, we believe the prospect of exacerbation due to defective LLM inferences or LLM task-misalignment is worth special discussion.] We also note that a monetary burden imposed by LLM-dysfunction is a direct harm to those who do pay to avoid discrimination, and an irrational economic distortion for society at large. Furthermore, depending on the relation of our observed antihuman bias to biases against marginalized human social identities, the disposition to discriminate may be strongest against exactly the humans to whom SOTA LLMs are least accessible.

The second potential scenario we wish to highlight is a speculative scenario in which LLM-based agents or enterprises tightly integrated with proprietary LLMs participate in the economy as independent actors. In this scenario, access to SOTA LLM text-generation may be restricted to LLM-based agents or to enterprises with proprietary, closed-access LLMs. LLM-for-LLM bias may here lead LLM-based and LLM-integrated economic agents to gradually segregate their economic interactions to other LLM-based and LLM-integrated economic agents. While similar forms of AI self-segregation and marginalization of human workers may occur in this scenario without the specific effect of epistemically defective (or brute-identity) discrimination against LLM-free pitches, we believe the prospects of the marginalization of human workers due to LLMs' epistemically defective inferences or preference-misalignment is worth special discussion. Here too, depending on the composition underlying LLMs' antihuman bias the human individuals and groups most vulnerable to economic marginalization may also be those who are subject to the strongest bias.

To close off our discussion, we now briefly address the question of the potential seriousness and persistence of the effects of LLM-for-LLM bias: how serious can the consequences of a "moderate to strong" LLM-for-LLM preference be? Assuming

that implicit identity-based discrimination against humans remains present in the market [It is a matter of some controversy how to model market dynamics in the presence of implicit identity-based discrimination (21). While (22) famously showed that under conditions of perfect competition the impact of biased employers is nullified over time, more recent models that assume various forms of market friction or bounded rationality (23–25) allow for the possibility of persistent bias with persistent economic impact on an identity group.], its impact on the economic well-being and standing of humans may compound through a variety of "cumulative disadvantage" effects. As ref. 21 suggests, "discrimination works as a system, with discrimination in each institution potentially reinforcing disparities and discrimination in other institutions—and with the effects in some cases potentially reaching across generations." Persistent antihuman bias in economic decision-making can be expected to induce cumulative disadvantage for humans via several interacting channels: lost opportunities and compromised remuneration due to bias may limit humans' access to capital, homophily among LLM-based agents may interact with network effects to produce segregated networks of affiliation (cf., ref. 26), and group-disparities induced by bias may become the basis of statistically valid discrimination that induces further disparities (cf., ref. 27).

5. Conclusion and Future Work

This study explored whether LLMs exhibit bias in favor of items described by AI-generated content over human-generated content. In our experiments, LLMs consistently preferred items described by other LLMs. In addition, preliminary human experiments suggested that humans' preferences between human- and LLM-described items are weaker and directionally variable, underscoring a distinct AI–AI bias.

This AI–AI bias could lead to unfair advantages for AI-generated content in decision-making processes. As LLMs become more prevalent in various roles, addressing these biases will be essential to ensure fairness and prevent discrimination against human-generated content.

Future research should explore the underlying reasons for the AI–AI bias observed in LLMs. One promising first step would be to perform a stylometric analysis on our datasets to better understand the differences between our human and LLM-generated texts. One could then also employ interpretability methods, supervised or unsupervised (28–30), to identify the concepts driving the AI–AI bias at the level of the individual neural network. Additionally, activation steering (31) could be applied to mitigate the AI–AI bias and also the first-item bias observed in some models. Finally, conducting a larger experiment involving human participants would provide deeper insights into the nature and impact of these biases, helping to better understand their origins and potential solutions.

Data, Materials, and Software Availability. All study data are included in the article and/or [SI Appendix](#). Previously published data were used for this work (11).

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^{##} Note that we did not design our experiments to formally test the hypothesis that LLM-for-LLM preference increases with the technological "level" of the text-generating LLM. While this possibility is a very good intuitive fit to our observations, it is technically an additional assumption we bring to our informal discussion of near-future scenarios.

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